MATH 202 Your name: _____

Learning target I3, version 1

Use integration by substitution and integration by parts to find antiderivatives. Decide which method to apply to a given integral.

Evaluate each of the following integrals. For each one:

- show all your work, including any new variables you use,
- and write one sentence that explains your choice of method and substitution.

Your process is even more important than your answer in each. Use the back of this paper if you need more room to explain your thinking.

1. $\int x^3 \cos(5x^4)\, dx$

2. $\int x \sin(5x)\, dx$

3. $\int \sin(x) e^{\cos(x)}\, dx$

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Learning target I3, version 2

Use integration by substitution and integration by parts to find antiderivatives. Decide which method to apply to a given integral.

Evaluate each of the following integrals. For each one:

- show all your work, including any new variables you use,
- and write one sentence that explains your choice of method and substitution.

Your process is even more important than your answer in each. Use the back of this paper if you need more room to explain your thinking.

1. $\int 5x\sin(2x)\,dx$

2. $\int e^x\cos(e^x)\,dx$

3. $\int x^7\ln(x)\,dx$

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Learning target I3, version 3

Use integration by substitution and integration by parts to find antiderivatives. Decide which method to apply to a given integral.

Evaluate each of the following integrals. For each one:

- show all your work, including any new variables you use,
- and write one sentence that explains your choice of method and substitution.

Your process is even more important than your answer in each. Use the back of this paper if you need more room to explain your thinking.

(a) $\int x^5 \cos(x^6+3)\,dx$

(b) $\int \theta^2 \sin(3\theta)\,d\theta$

(c) $\int \dfrac{7x^2}{\sqrt{4x^3+11.2534}}\,dx$

MATH 202 Your name: _____

Learning target I3, version 4

Use integration by substitution and integration by parts to find antiderivatives. Decide which method to apply to a given integral.

Evaluate each of the following integrals. For each one:

- show all your work, including any new variables you use,
- and write one sentence that explains your choice of method and substitution.

Your process is even more important than your answer in each. Use the back of this paper if you need more room to explain your thinking.

(a) $\int x \ln(x)\, dx$

(b) $\int y^5 \sin(3y^6 - 2)\, dy$

(c) $\int 5q e^{-5q}\, dq$